

Senior Ontology - Smart Route & User Management API

Unit test specification: Login (2.0)

Scope note: Backend tests target server.py (FastAPI). DFD processes 2.1–2.2 and 2.0 are mapped to POST /login and mocked User Info store. Flutter UI validation is out of scope for these Python unit tests.

Test suite B — Process 2.0 Login

Test Case ID	TC-LOGIN-2.0
Python module	UnitTest_2_0_Login.py
Priority	High
Data store	User Info (mocked)

DFD sub-processes covered

- 2.1 Get user information — Email and password submitted as LoginRequest.
 - 2.2 Check existing user — Server scans store; match on email (case-insensitive) and exact password.
 - 2.0 Login completion — Returns User Info subset (firstname, email, etc.) to client on success.
- Supporting: root/health availability — test_api_health_root_available.

Step	Action	Expected Result	Status
1	POST /login valid credentials	200, Success + user	Pass
2	POST /login case-insensitive email	200, Success	Pass
3	POST /login wrong password	401, Invalid credentials	Pass
4	POST /login unknown email	401, Invalid credentials	Pass
5	GET root / health (supporting)	message present; ok True	Pass

DFD / requirement	Unit test method
2.1–2.2 Input + validation	test_dfd_2_0_login_success_returns_user_info, test_dfd_2_2_case_insensitive_email
2.2 Failure paths	test_dfd_2_2_wrong_password, test_dfd_2_2_unknown_email
2.0 Success response	test_dfd_2_0_login_success_returns_user_info, test_dfd_2_2_case_insensitive_email
Service availability	test_api_health_root_available

Related (post-registration): Change password flows — see 1.0 suite / test_dfd_1_3_record_and_post_registration_password.